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Batch :- B3

ASSIGNMENT No: 1

TITLE: Develop an application based on Java environment setup, JDK/JRE/JVM, variables, methods and constructors. and demonstrate the implementation of the concepts through suitable examples.

Problem Statement

Develop an application based on Java environment setup, JDK/JRE/JVM, variables, methods and constructors. and demonstrate the implementation of the concepts through suitable examples.

Learning Objectives

Understand the theoretical concepts, implement them in Java programs, analyze outputs, and apply the concepts to solve practical programming problems.

Theory

Java environment setup, JDK/JRE/JVM, variables, methods and constructors. forms an important component of Java programming. Students should understand the underlying concepts, architecture, advantages, use cases, and implementation methodology. The exercise should focus on understanding the design principles, coding practices, execution flow, and practical applications of these concepts in software development. Detailed study should include syntax, APIs, execution process, and real-world applications.

Code and Output :-

```
public class Main
{
    // Method
    static int square (int number)
    {
        return number*number;
    }
    public static void main(String[] args)
    {
        // Variables
        String name = "Parth";
        int age = 19;
        double salary = 55000.50;
        int result = square(6);
        System.out.println("Name :" +name);
        System.out.println("Age :" +age);
        System.out.println("Salary :" +salary);
        System.out.println("square of 6 = |" +result);
    }
}
```



```
C:\Users\CL408_07\Desktop\Parth java>javac main.java
```

```
C:\Users\CL408_07\Desktop\Parth java>java Main
```

```
Name :Parth
```

```
Age :19
```

```
Salary :55000.5
```

```
public class Main
{
    // Method
    static int square(int number)
    {
        return number * number;
    }

    public static void main(String[] args)
    {

        String name = "Parth";
        int age = 19;
        double salary = 55000.50;
        int result = square(6);
        System.out.println("Name :" + name);
        System.out.println("Age :" + age);
        System.out.println("Salary :" + salary);
        System.out.println("square of 6 = " + result);

        int num1 = Integer.parseInt(args[0]);
        int num2 = Integer.parseInt(args[1]);

        int sum = num1 + num2;
        int difference = num1 - num2;
        int product = num1 * num2;
        float quotient = (float) num1 / num2;

        System.out.println("Addition: " + sum);
        System.out.println("Subtraction: " + difference);
        System.out.println("Multiplication: " + product);
        System.out.println("Division: " + quotient);
    }
}
```

```
D:\Parth Java>javac Main.java
```

```
D:\Parth Java>java Main 10 5
```

```
Name :Parth
```

```
Age :19
```

```
Salary :55000.5
```

```
square of 6 = 36
```

```
Addition: 15
```

```
Subtraction: 5
```

```
Multiplication: 50
```

```
Division: 2.0
```



Exercise

1. Explain the main concept used in Java environment setup, variables and methods.
2. What are the advantages of using Java variables and methods?
3. How can this implementation be improved?
4. What are the real-world applications?

Answers :-

1 . The **Java environment setup** is the process of installing the **Java Development Kit (JDK)** and configuring the **PATH** environment variable so that Java programs can be compiled and executed. **Variables** are named memory locations used to store different types of data such as numbers and text. **Methods** are reusable blocks of code that perform specific tasks, improve code organization, reduce repetition, and make programs easier to understand and maintain.

2. Advantages of Variables

- Store data efficiently.
- Make programs flexible and dynamic.
- Improve readability.
- Reduce repetition of values.
- Allow easy modification of data.

Advantages of Methods

- Promote code reusability.
- Reduce duplicate code.
- Improve program modularity.
- Make debugging and maintenance easier.
- Increase readability and organization.

3. The implementation can be improved by:

- Using meaningful variable and method names.
- Following Java naming conventions.
- Dividing large programs into smaller methods.
- Handling exceptions properly.
- Adding comments for better understanding.
- Validating user input.
- Using object-oriented concepts like classes and objects.



- Keeping code clean and well-formatted.

4. Java variables and methods are widely used in many applications, including:

- **Banking Systems** – Managing customer accounts, balances, and transactions.
- **Student Management Systems** – Storing student information and calculating results.
- **E-Commerce Applications** – Managing products, customers, and orders.
- **Hospital Management Systems** – Maintaining patient records and appointments.
- **Online Food Delivery Apps** – Processing orders and payments.
- **Library Management Systems** – Managing books and member details.
- **Inventory Management Systems** – Tracking stock and sales.
- **Mobile Applications (Android)** – Building Android apps using Java.
- **Enterprise Software** – Developing secure business applications.
- **Game Development** – Creating simple Java-based games.